It is also called an absolute value function.

Remark Domain is R and range set is $[0, \infty)$

NOTE

$|5| = 5, |-5| = -(-5) = 5$

5.1.13 Signum function

A function $f(x)$ is defined by

$f(x) = \begin{cases} \frac{|x|}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ is called the Signum function.

Remark Domain is R and range set is $\{-1, 0, 1\}$

5.1.14 Step function

(i) Greatest integer function

The function whose value at any real number x is the greatest integer less than or equal to x is called the greatest integer function. It is denoted by $\lfloor x \rfloor$.

i.e. $f : R \longrightarrow R$ defined by $f(x) = \lfloor x \rfloor$ is called the greatest integer function.

NOTE

$\lfloor 2.5 \rfloor = 2,\ \lfloor -2.1 \rfloor = -3,$
$\lfloor 0.74 \rfloor = 0,\ \lfloor -0.3 \rfloor = -1, \lfloor 4 \rfloor = 4.$

(ii) Least integer function

The function whose value at any real number x is the smallest integer greater than or equal to x is called the least integer function. It is denoted by $\lceil x \rceil$.

i.e. $f : R \longrightarrow R$ defined by $f(x) = \lceil x \rceil$ is called the least integer function.

Remark Function's domain is R and range is Z [set of integers]

NOTE

$\lceil 4.7 \rceil = 5, \lceil -7.2 \rceil = -7, \lceil 5 \rceil = 5, \lceil 0.75 \rceil = 1.$

5.1.15 Rational function

A function $f(x)$ is defined by the rule $f(x) = \frac{p(x)}{q(x)}$, $q(x) \neq 0$ is called rational function.

Example: $f(x) = \frac{x^2 + 1}{x - 3}, x \neq 3$ is a rational function.

5.1.16 Polynomial function

For the real numbers $a_0, a_1, a_2, \ldots, a_n$; $a_0 \neq 0$ and n is a non-negative integer, a function $f(x)$ given by $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \ldots + a_n$ is called as a polynomial function of degree n.

Example: $f(x) = 2x^3 + 3x^2 + 2x - 7$ is a polynomial function of degree 3.

5.1.17 Linear function

For the real numbers a and b with $a \neq 0$, a function $f(x) = ax + b$ is called a linear function.

Example: $y = 2x + 3$ is a linear function.

5.1.18 Quadratic function

For the real numbers a, b and c with $a \neq 0$, a function $f(x) = ax^2 + bx + c$ is called a quadratic function.

Example: $f(x) = 3x^2 + 2x - 7$ is a quadratic function.

5.1.19 Exponential function

A function $f(x) = a^x$, $a \neq 1$ and $a > 0$, for all $x \in R$ is called an exponential function

Remark Domain is R and range is $(0, \infty)$ and $(0, 1)$ is a point on the graph

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